

CS1060 HW2: Design Areas

Collaborators: Katherine Harvey and Waseem Ahmad

Product: HUDS Nutrition Analyzer

Goal: To help students make healthier and more informed decisions about their food choices in the dining hall by uploading a photo, tracking their nutrition intake, and providing personalized recommendations.

Similar Existing Services:

- MyFitnessPal
- Cronometer
- YAZIO: AI Calorie Tracker

Stakeholders:

- Direct:
 - Harvard College students (athletes, health conscious individuals, etc)
 - Dining hall staff
 - HUDS
 - Development team (Katherine, Waseem, and any future group members)
- Indirect:
 - Parents
 - Healthcare providers
 - Other schools and universities
 - Investors

Differentiation:

While apps like MyFitnessPal, Cronometer, and YAZIO offer general food logging and nutrition tracking, they're not designed for Harvard dining halls and often lack data about dining hall meals, which makes it quite tedious to log and keep track of your daily food intake. Our nutrition analyzer stands out in the following ways. Note that this is just a general idea as of now, so the final product may have fewer (or more) features than listed below.

- Uses HUDS menu items, portion sizes, and plating styles to generally give more accurate nutrition info from a photo
- Students can create an account which lets the app remember preferences, allergies, and goals without requiring access to sensitive Harvard ID data.
- Automatically loads the daily HUDS menus and hours, so students can plan meals ahead of time and see what fits their goals before arriving.
- Based on what students eat over time, we could highlight nutrient gaps and suggest upcoming menu items to help them make healthier choices.
- Add features like streaks, house affiliation, and crowd level reports, where students can report how crowded a dining hall is during breakfast/lunch/dinner.

- House affiliation could be used to help students find the closest dining hall to their current location after taking interhouse restrictions into consideration.

User Journey:

User

Healthy Hannah			Expectations		
Hannah wants to eat better at her dining hall without overthinking her nutrition. She wants quick and accurate information about her plate and recommendations to reach her goals.			• Fast, reliable nutrition analysis from a photo • Easy-to-understand feedback and recommendations • Easy progress tracking		
	Define	Capture and Analyze	Recommend	Track	Sustain
Steps	(1) Learns about the app via dining hall poster (2) Downloads app and signs up (3) Set dietary preferences, allergies, and nutrition goals	(4) Opens app and snaps photo of plate in the dining hall (5) App detects food and portion sizes (6) Gets nutrition facts instantly (calories, macros, key vitamins)	(7) App highlights nutrient gaps (8) Provides gentle suggestions	(9) Dashboard shows daily totals and weekly nutrient balance (10) User views streaks, progress toward goals, and achievements	(11) Receives weekly insights (12) Gets personalized recommendations for dining hall
User Thoughts	"I'd love to know what I'm really eating"	"That's more carbs than I expected"	"Okay, I'll grab some fruit to round this out"	"Nice, I've been consistent with protein all week"	"This app is helping me find better choices in the dhall"

Opportunities	• Simple onboarding • Integration with dining hall schedule	• Improve accuracy of portion recognition • Allow quick correction / ask the user if the estimate seems right	• Encourage positive reinforcement • Tailor nudges to goals	• Gamification (streaks, badges, challenges)	• Push notifications tied to dining hall menu • Community or house challenges
Internal Ownership + Metrics • Tech Team: Improve food recognition accuracy and speed • UX Team: Make nutrition feedback clear and easy to understand (>80% positive rating) • Nutrition Team: Ensure recommendations align with dietary guidelines • Marketing Team: Track adoption and weekly use					

Value Tensions:

Product Tech Team

- The tech team wants accurate, reliable data and might design more complex steps, while Hannah just wants quick, low-effort logging and might skip steps that slow her down.

Other Harvard Students

- Other students may treat the app as a competition (streaks, leaderboards), while Hannah just wants personal feedback and might feel pressured or judged by comparison.

HUDS

- HUDS may want students to try a range of foods so the app can provide useful data about menu performance, while Hannah may only log her favorite meals, giving an incomplete picture.

Dining hall staff

- Hannah might want more variety or specific nutrients, but dining staff can only serve the standard menu and may not be able to accommodate her preferences.

Healthcare Providers

- Health staff want to encourage long-term balanced eating habits, while Hannah mainly wants quick day-to-day feedback and may not want to think about long-term goals.

Investor

Investor Ian			Expectations		
Ian is an alum who casually donates to student projects. He likes supporting things that clearly improve student life, but he doesn't want something insanely complex or unintuitive, just something simple that's quick and works well.			• Simple value proposition • Evidence of adoption and retention • Easy to explain impact • Low risk and clear use by students		
	Define	Capture and Analyze	Recommend	Track	Sustain
Steps	(1) Sees the HUDS Nutrition Analyzer mentioned in a campus newsletter (2) Reaches out to the development team for a demo video	(3) Learns it uses photos + Harvard menus to give fast nutrition info (4) Sees students are already testing it and reads about accuracy	(5) Asks the team to keep the design simple and student-friendly (6) Suggests showing daily HUDS menus in a clean, easy format so students know what's offered (7) Recommends highlighting how the app can save time by showing which hall is least crowded	(8) Gets monthly updates showing growing student usage and positive feedback	(9) Provides a grant to help the team keep improving and promoting the app on campus (10) Shares the project with the Harvard Alumni Association newsletter to attract more support
Investor Thoughts	"This seems useful and solves a real student need."	"It looks simple enough that students will actually use it, and it already has	"Let's keep it simple so more students actually use it, and highlight how	"This is actually improving student health, and people are actually using	"I'm glad I supported this, and perhaps with the grant it could be maintained

		some traction.”	it can save time as well.”	it to track their meals.”	for a while and scale beyond Harvard.”
Opportunities	Create a short, visual demo that explains the value in under a minute	Show quotes from early testers and screenshots of the interface	Emphasize speed and simplicity in the UI; start with a small pilot group	Send short, visual update emails with a few key stats and student comments	Provide a thank-you summary with student testimonials and impact visuals
Internal Ownership + Metrics • Tech Team: Ensure app loads and analyzes photos quickly (ideally <5s) • UX Team: >80% of users say it’s “easy to use” • Nutrition Team: Highlight simple, actionable recommendations • Marketing Team: Show weekly active user growth					

Value Tensions:

Product Development Team

- Ian wants something simple and quick to build, while the development team may want to add more features and testing to make it sustainable over time.

Harvard Students

- Ian wants lots of students using it quickly to show success, while students may prefer using it casually and at their own pace without pressure.

Dining hall staff

- Ian wants fast, visible results, while HUDS staff want to ensure the rollout is organized and doesn’t add extra work or confusion in dining halls.

Marketing Team

- Ian wants to keep the app low-profile and simple, while the marketing team may want to heavily promote it to drive engagement.